

Silicon Photonic Interferometer Design

Design proposal and simulation results

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I Introduction

I designed six TE devices to measure the waveguide group index from their free spectral range (FSR). The layout contains two Mach–Zehnder interferometers (MZIs), two Fabry–Pérot (FP) cavities, a Michelson interferometer, and a Michelson interferometer with an FP cavity in one arm, which I refer to as MFP. I chose different path lengths so that I could compare the fringe spacing and examine the effect of the Bragg mirrors.

The saved INTERCONNECT spectra give a group index of about 4.19 for both MZIs and the Michelson. The FP cavities have smaller FSRs than the estimate based on waveguide length alone. Their common reflection delay can be examined by comparing the two cavity lengths.

Table I Devices in EBeam_weizhu.gds

Cell name	Length in the submitted layout
TE_MZI_30um	Arm lengths 150.001 and 179.999 μm ; difference 29.998 μm
TE_MZI_150um	Arm lengths 149.999 and 300.001 μm ; difference 150.002 μm
TE_FP_150um	Cavity waveguide 149.999 μm
TE_FP_250um	Cavity waveguide 249.999 μm
TE_MM_100um	Michelson arm difference 100 μm
TE_MFP_100um	FP cavity waveguide 100 μm in one reflecting arm

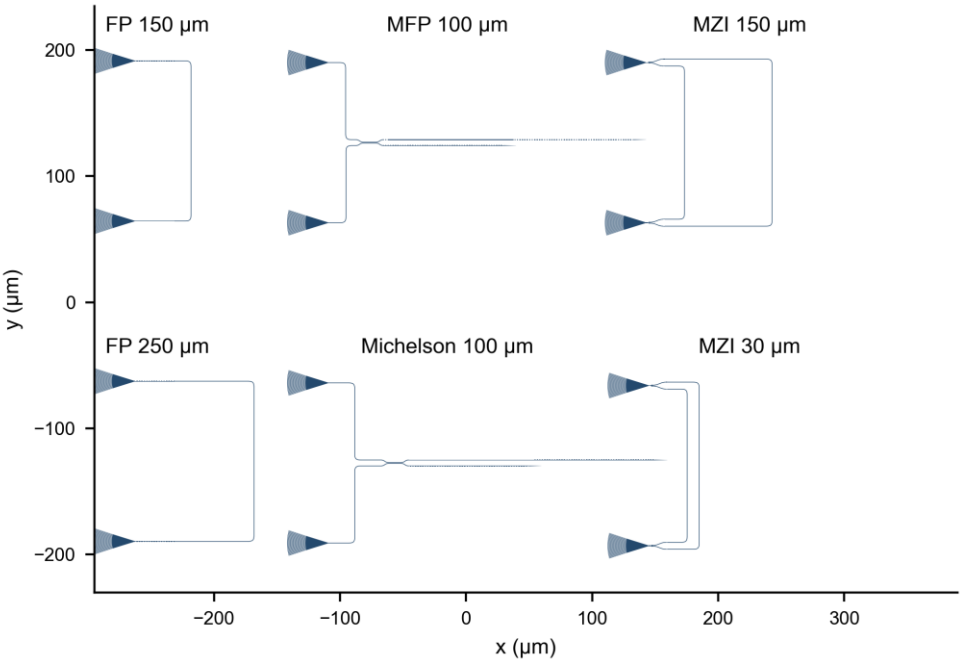


Figure 1 Silicon geometry from the submitted layout, top cell EBeam_weizhu_v1. Each device has one input and one output grating coupler. The vertical coupler spacing is 127 μm .

II Theory

The layout uses the EBeam PDK strip waveguide with a nominal width of 500 nm and silicon thickness of 220 nm. The TE PDK model gives an effective index of 2.44553 and group index of 4.19088 at 1550 nm [1]. I use the group index as the reference for the analytical FSR estimates.

The propagation constant β and group index are related to the wavelength-dependent effective index by

$$\beta = \frac{2\pi n_{eff}}{\lambda}, \quad n_g = n_{eff} - \lambda \frac{dn_{eff}}{d\lambda} \quad (1)$$

For an ideal MZI with equal splitting and equal arm loss, the normalized transmission is

$$T_{MZI} = \frac{1 + \cos(\beta \Delta L)}{2} \quad (2)$$

ΔL is the difference between the arm lengths. The FSR in frequency and the approximate spacing in wavelength are

$$\Delta f_{MZI} = \frac{c}{n_g \Delta L}, \quad \Delta \lambda_{MZI} \approx \frac{\lambda^2}{n_g \Delta L} \quad (3)$$

In the Michelson, light travels to the mirrors and back, so the arm difference is doubled. For identical mirrors with equal reflection phase,

$$\Delta f_{MM} = \frac{c}{2n_g \Delta L}, \quad \Delta \lambda_{MM} \approx \frac{\lambda^2}{2n_g \Delta L} \quad (4)$$

For an FP cavity, the full round-trip delay includes propagation through the cavity and reflection from both gratings. If the mirror reflection delays are τ_1 and τ_2 ,

$$\Delta f_{FP} = \frac{1}{\tau_{rt}}, \quad \tau_{rt} = \frac{2n_g L}{c} + \tau_1 + \tau_2 \quad (5)$$

Here L is the waveguide length between the grating interfaces and c is the speed of light in vacuum. Omitting mirror delay gives FP estimates of 1.91 nm for 150 μm and 1.15 nm for 250 μm at 1550 nm. The same calculation gives 19.11 and 3.82 nm for the two MZIs, and 2.87 nm for the 100 μm Michelson.

The MFP output combines the field reflected by the cavity with the field reflected by the reference arm. Its output fringe spacing therefore depends on both the cavity reflection phase and the arm phase difference. I keep its measured output spacing separate from the group-index calculation.

III Modelling and simulation

I simulated the circuits in Lumerical INTERCONNECT and saved one text export for each device on 5 October 2026. The six files contain 5,000 wavelength and Y pairs over 1499.02–1601.00 nm, with about 0.02 nm wavelength spacing near 1550 nm. Figure 2 shows the exported gain traces without smoothing or normalization.

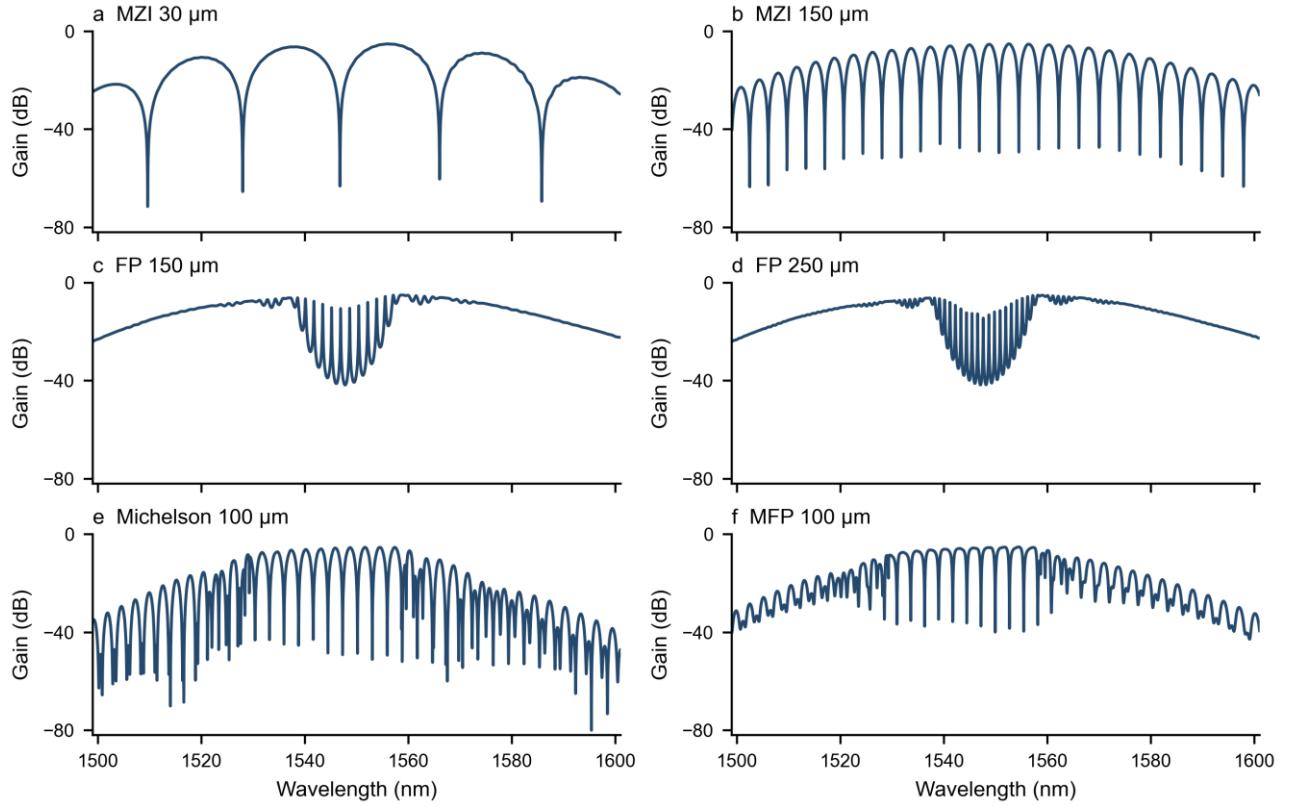


Figure 2 Full saved INTERCONNECT spectra for the six devices. Panels a and b show the MZIs, c and d the FP cavities, e the Michelson, and f the MFP. The gain values include the grating couplers and other circuit components.

All Bragg gratings in the submitted GDS have a period of 317 nm. The standalone FP cavities use 100-period mirrors with 50 nm corrugation. The Michelson has two 300-period mirrors with 200 nm corrugation. The MFP has a 10-period front mirror, a 300-period rear mirror, and a 300-period reference mirror, all with 200 nm corrugation. The short front mirror is intended to couple light into the FP cavity.

The MZIs use Y branches as the splitter and combiner. The Michelson and MFP use the EBeam directional coupler, with a 500 nm waveguide width, 200 nm gap, 10 μm coupling length, and 5 μm radius. The routed waveguides also have a 5 μm bend radius. These dimensions are taken from the GDS metadata.

For comparison, the earlier waveguide export wg.mat contains ten samples of effective index and group velocity. Linear interpolation of those data gives an effective index of 2.4462 and group index of 4.2036 at 1550 nm. This is a separate waveguide result. The reference value used for the circuit FSR estimates is the PDK group index of 4.19088.

Data sources MZI layout test folder: TE_MZI_30um.txt, TE_MZI_150um.txt, TE_FP_150.txt, TE_FP_250.txt, TE_MM_100um.txt, and TE_MFP_100um.txt. The exports contain the spectrum values but not the full circuit setup.

Simulated free spectral range

I used consecutive minima for the MZIs and reflecting interferometers, and transmission maxima for the FP cavities. Figure 3 marks the selected features. Most are in the 1542–1555 nm window. The 30 μm MZI needs a wider window because its FSR is about 19 nm.

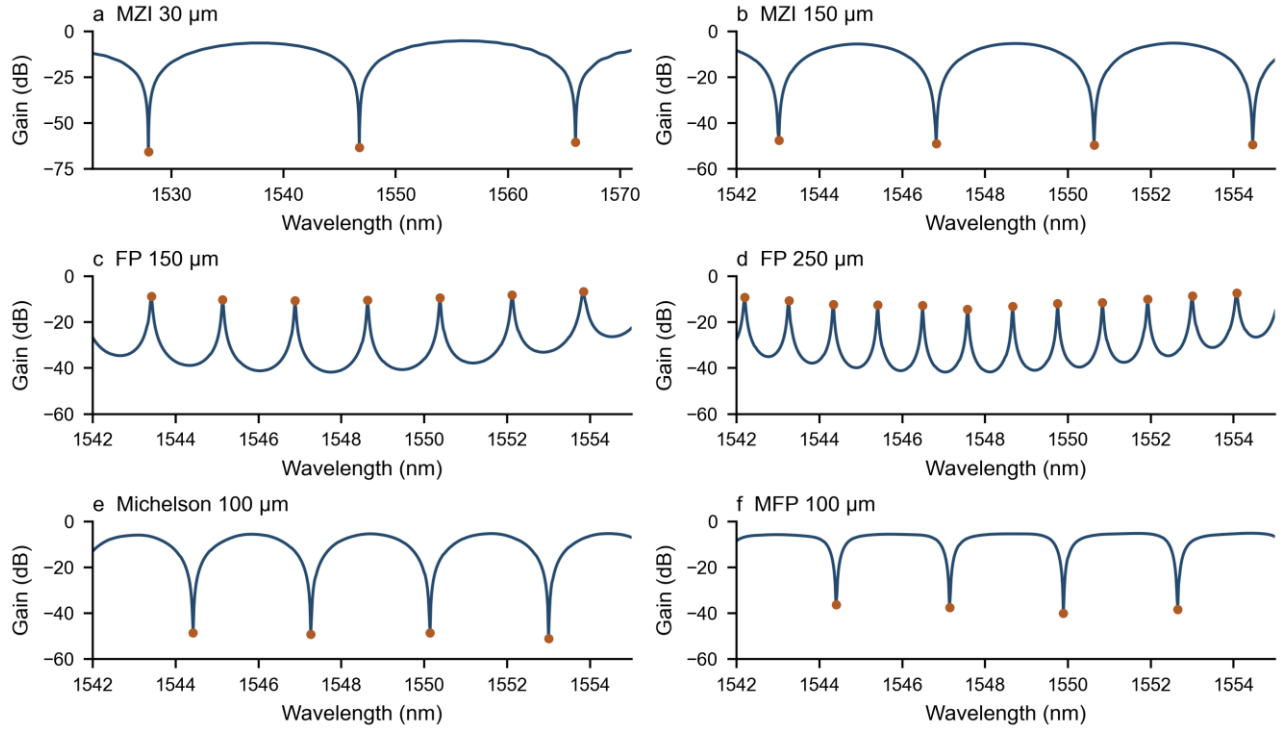


Figure 3 Spectral windows used for the FSR calculation. Orange markers identify the selected features. Panel a uses a wider wavelength range than the other panels.

Table II Analytical and simulated FSR values

Device	Analytical FSR ¹	Saved spectrum FSR	Frequency FSR	Group index ²
MZI 30 μm	19.11 nm	19.03 nm	2384.3 GHz	4.192
MZI 150 μm	3.82 nm	3.81 nm	476.6 GHz	4.193
Michelson 100 μm	2.87 nm	2.86 nm	357.5 GHz	4.193
FP 150 μm	1.91 nm	1.74 nm	217.3 GHz	4.599 effective
FP 250 μm	1.15 nm	1.08 nm	135.1 GHz	4.438 effective
MFP output	—	2.75 nm	343.8 GHz	—

¹ Estimates at 1550 nm using the PDK group index. FP estimates omit mirror delay. ² Individual FP values are effective cavity indices including mirror delay. No intrinsic group index is assigned to the MFP output.

I converted each selected wavelength to frequency using $f = c/\lambda$ and averaged the adjacent frequency intervals. Rearranging Equations 3 and 4 gives the MZI and Michelson group indices. The selected minima for the 30 μm MZI are 1527.95, 1546.75, and 1566.01 nm. The 150 μm MZI uses 1543.02, 1546.82, 1550.63, and 1554.46 nm. All three extracted indices agree with the PDK value to about 0.06%.

The Michelson minima at 1544.42, 1547.26, 1550.14, and 1553.00 nm are regularly spaced near the band centre. More complicated features appear outside this region in the Michelson and MFP traces. I have not assigned a single fringe family to the entire sweep. The finite wavelength sampling also limits the accuracy of narrow FP peak heights.

IV Fabrication

The design uses the EBeam PDK components for the 220 nm silicon platform. I placed 12 TE grating couplers in six input–output pairs at 127 μm pitch. The MZI layout uses one input and one output, with the Y branches connecting the two internal arms. Four EBeam terminators are placed at the unused far ends of the Michelson and MFP gratings.

The GDS contains 78 optical pins at 39 matching locations. The layout file matches the submitted EBeam_weizhu.gds, which was merged in pull request 370 on 4 October 2026 [2]. The automated submission checks passed. Fabricated-device measurements are not available yet.

V Experimental data

After fabrication, I plan to record the output power while sweeping a TE-polarized laser near 1550 nm. I will use several neighbouring fringes to calculate the FSR in frequency, then compare the MZI and Michelson group indices. The values in Table II are calculated from simulation exports.

VI Analysis

The two FP cavities have the same mirror design but differ in cavity length by 100 μm . Their simulated round-trip delays are 4.602 ps and 7.402 ps. Assuming the common Bragg reflection delay is approximately equal over the selected central windows, subtracting these delays removes the mirror contribution to first order:

$$n_g \approx \frac{c}{2(L_{250} - L_{150})} \left[\frac{1}{\Delta f_{250}} - \frac{1}{\Delta f_{150}} \right] \quad (6)$$

This gives a group index of 4.198, close to the MZI and Michelson values. Using the PDK group index, the individual FP spacings correspond to effective cavity lengths of 164.6 and 264.8 μm . Both are about 15 μm longer than the waveguide between the gratings, which is consistent with an additional reflection delay. The MFP output spacing is 2.75 nm. Its cavity and reference-arm phases must be separated before using it to estimate a waveguide index.

VII Conclusion

The MZI and Michelson results give a consistent group index near 4.19. Increasing the path difference reduces the FSR as expected. The paired FP calculation gives a similar result after accounting for the common mirror delay. The MFP has regular central fringes, but its combined reflection response needs a separate phase analysis.

References

- [1] SiEPIC, SiEPIC EBeam PDK, installed version 0.4.53. Strip TE waveguide model for $w = 500$ nm and $h = 220$ nm. https://github.com/SiEPIC/SiEPIC_EBeam_PDK
- [2] Wei Zhu, EBeam_weizhu.gds, openEBL October 2026 submission, pull request 370. <https://github.com/SiEPIC/openEBL-2026-10/pull/370>
- [3] Ansys Optics, Optical Network Analyzer ONA INTERCONNECT Element. <https://optics.ansys.com/hc/en-us/articles/360036617973>
- [4] Ansys Optics, Bragg Grating WBG INTERCONNECT Element. <https://optics.ansys.com/hc/en-us/articles/360036108114>